23CSE202 DATABASE MANAGEMENT SYSTEMS
S3 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

2. UI Design Document

Group No : **A8**

Team Members :

Roll No	Student Name
AM.SC.U4CSE25014	ADITHYAN B
AM.SC.U4CSE25040	MADHAV A
AM.SC.U4CSE25024	ALI ADNAN THAHA
AM.SC.U4CSE25049	ROBIN ANTONY

Project Title : **Smart Digital Library Management System**

2.1. Functional requirements:

1. Secure role-based login for two user types: Library Administrator and Student.

2. Book catalog search and filtering by title, author, category, and availability.

3. Book issue and return management, including due-date assignment and status tracking.

4. Reservation handling for books that are currently unavailable, with admin approve/reject actions.

5. Automatic overdue detection and fine calculation for late returns.

6. Add, edit, and delete book records (CRUD) with accession number and copy-count tracking.

7. Student record management, including active loan count and outstanding fines per student.

8. Borrowing history and fine history view for each student.

9. Report generation and analytics: circulation trends, most-borrowed categories, and collection status.

10. Real-time dashboard summarising total titles, active loans, overdue count, and pending fines.

2.2. Non functional requirements:

1. Usability: minimal-click workflows for issuing, returning, and reserving books, with immediate visual confirmation.

2. Responsiveness: layout adapts cleanly from desktop down to tablet and mobile widths.

3. Performance: catalog search and filtering respond instantly as the user types (client-side filtering).

4. Consistency: a single design system (colour, type, spacing) is used across every screen and role.

5. Data integrity: forms validate required fields (e.g. title, author) before records are saved.

6. Accessibility: visible focus states, adequate colour contrast, and readable type scale throughout.

7. Scalability: the data-driven rendering approach supports a growing catalog and user base without UI changes.

8. Maintainability: HTML structure, styling, and logic are separated into index.html, styles.css, and app.js.

2.3. List of AI tools used for building UI(if any):

1. Claude (Anthropic) — used to design and generate the complete interactive HTML/CSS/JS UI, including the visual design system, all screens for both roles, and the sample interaction data.

2. Google Fonts (Fraunces, Inter, IBM Plex Mono) — used within the design for typography, integrated via a Claude-generated stylesheet.

2.4. Prompt (if any) :

Sl. No	Name of the tool	prompts
1	Claude (Anthropic) — follow-up design direction	Design the interface around the Smart Digital Library Management System abstract: separate Administrator and Student roles, a card-catalog inspired visual identity, flip-card book search results, and a due-date 'stamp' animation for issue/return/approve actions, built as plain HTML/CSS/JS with Chart.js for the reports dashboard.

2.5. List of files:

The following files were created for the UI and are included in the submitted GitHub repository:

index.html	Page structure for the login screen and the full Admin / Student application shell (all views, modal, forms).
styles.css	Complete visual design system — colour tokens, typography, card-catalog styling, flip-card and stamp animations, responsive layout.
app.js	Application logic — in-memory data model for books/students/loans/reservations/fines, view routing, CRUD actions, Chart.js dashboards.

2.6. UI Screenshot:

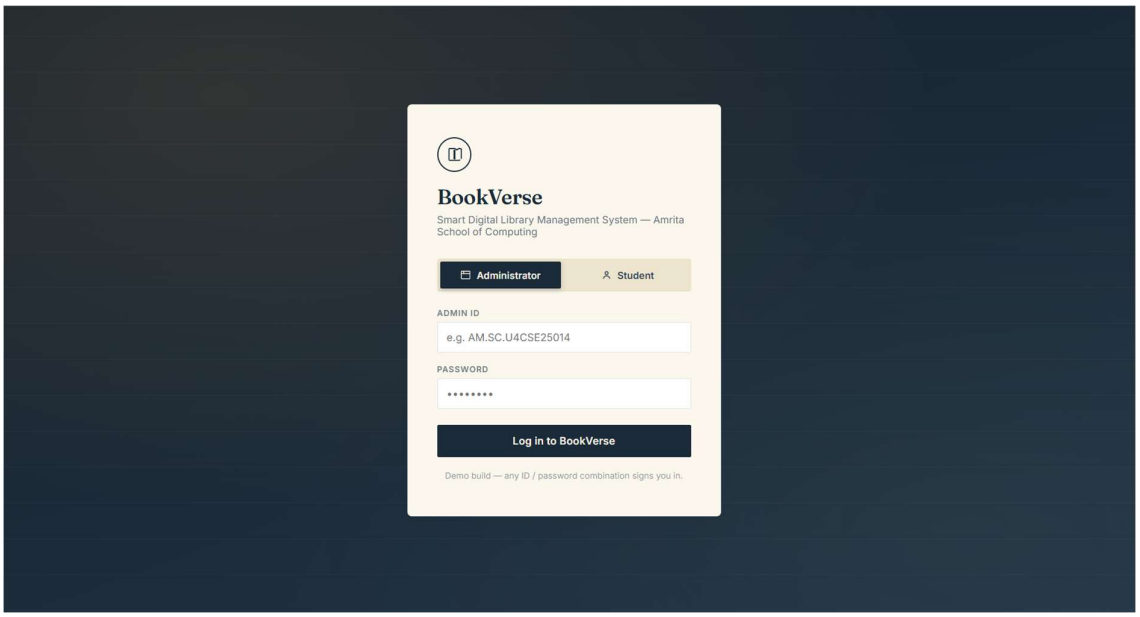


Fig 1. Role-based login — Administrator / Student

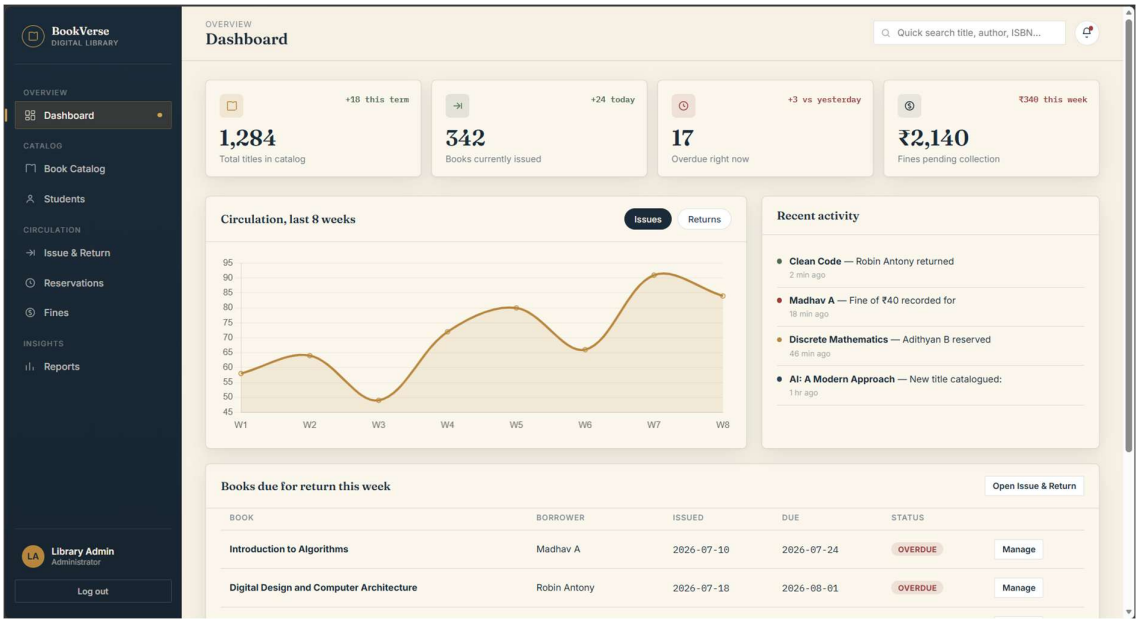


Fig 2. Administrator dashboard — live stats, circulation chart, due-soon table

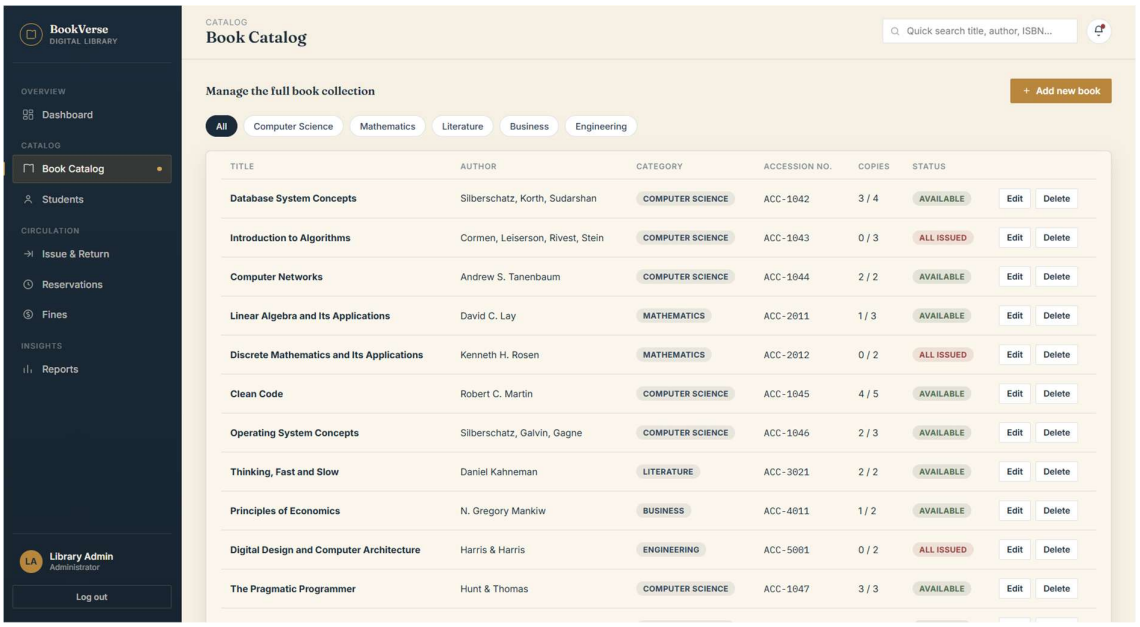


Fig 3. Administrator — book catalog management (CRUD)

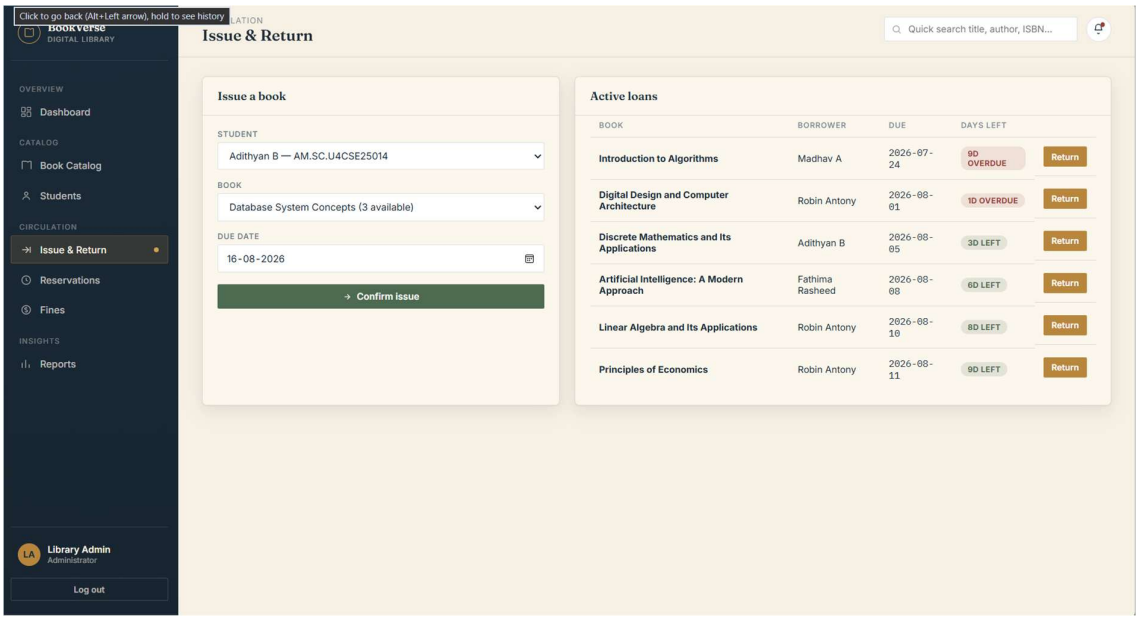


Fig 4. Administrator — Issue & Return workflow with active loans

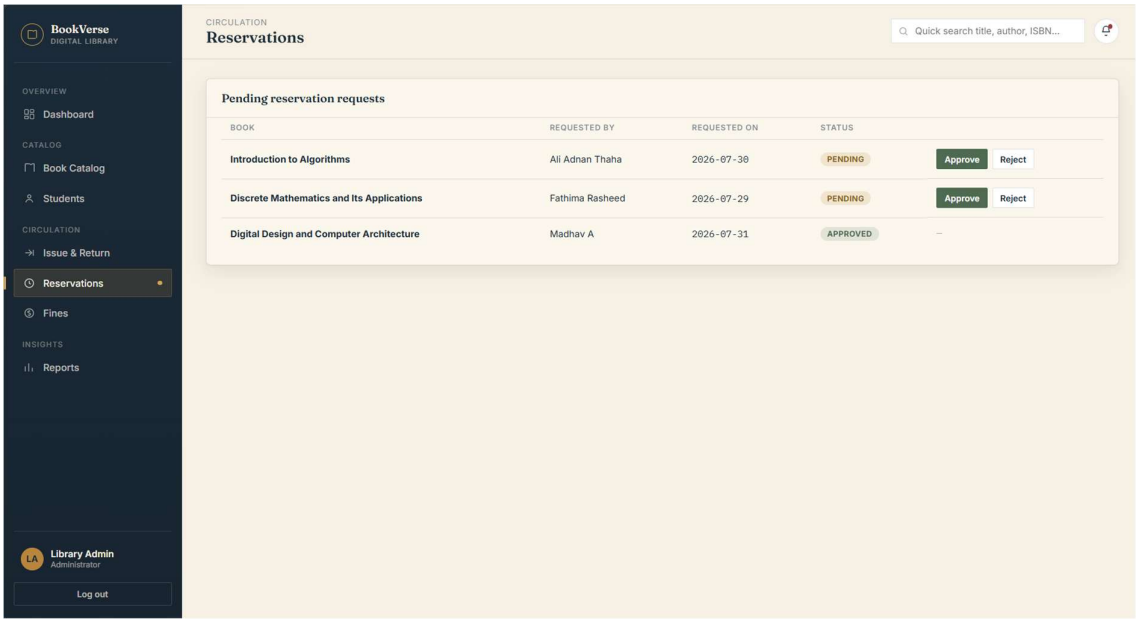


Fig 5. Administrator — reservation approval queue



Fig 6. Administrator — reports

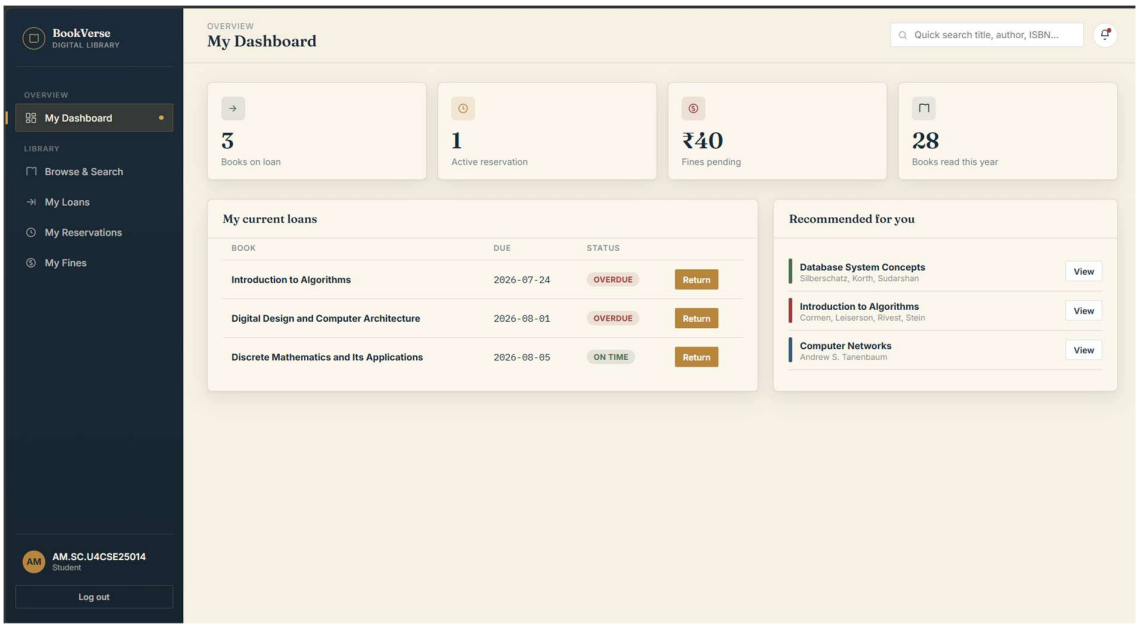


Fig 7. Student Dashboard

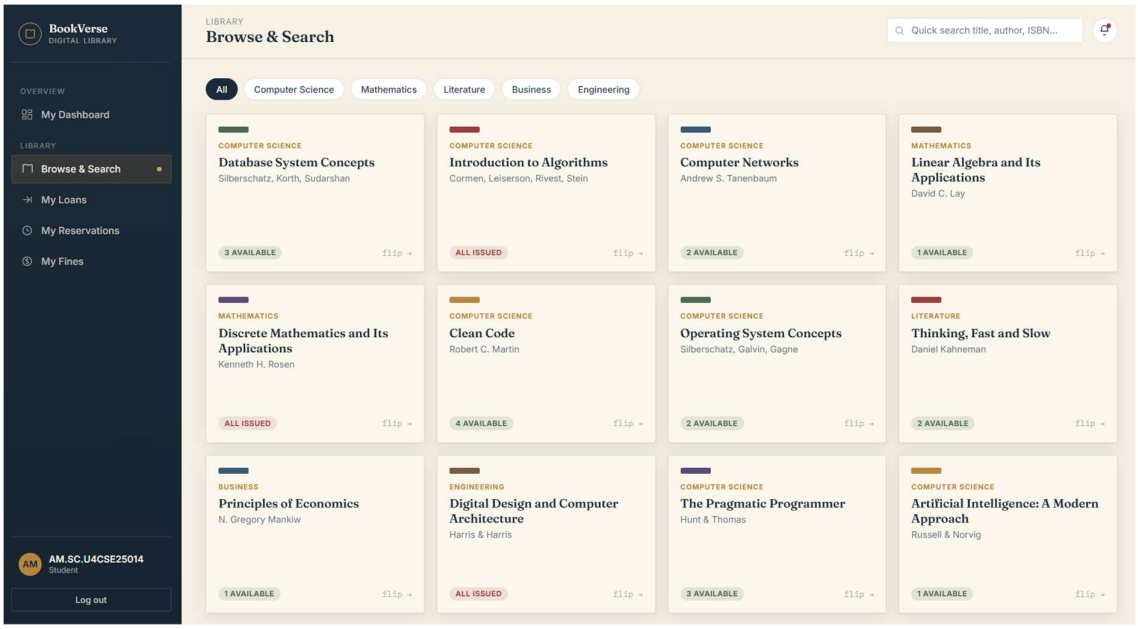


Fig 8 . Student — book catalog as flip-able index cards

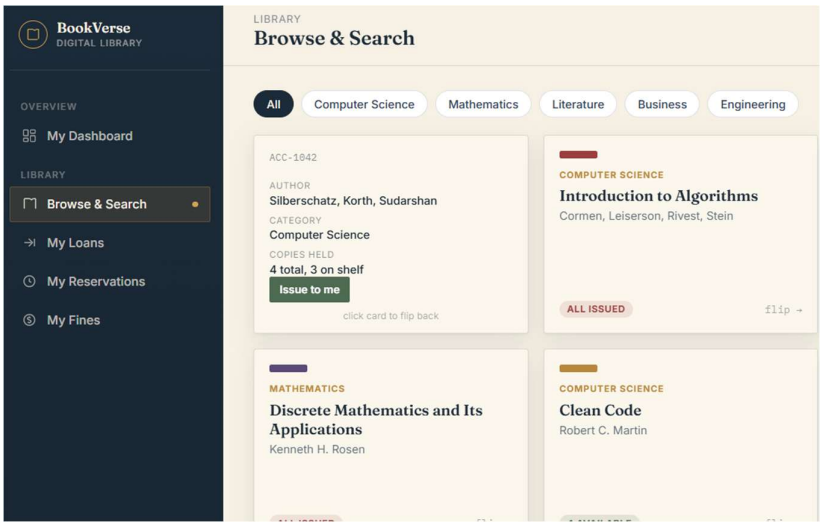


Fig 9. Student — index card flipped to show accession details and Reserve/Issue action

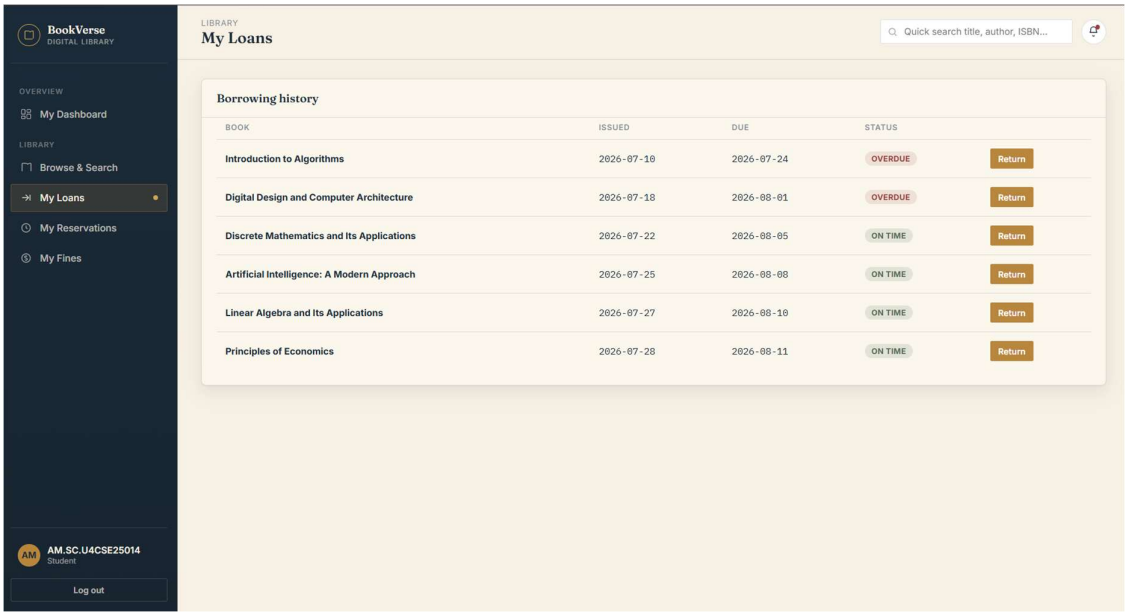


Fig 10. Student — borrow history

2.7. GitHub Link:

Github link : https://github.com/adithyanbalu/BookVerse-SmartLibrary

After deploying website live link :

https://adithyanbalu.github.io/BookVerse-SmartLibrary/